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SIMATS ENGINEERING
ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Assessment Tool Weightage: 40%

Total Marks:50

QUESTIONS: 10

Annexure A
Short Answer Test

Code of Conduct:

I Vignesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Short Answer Test
Questions

82/100

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.
2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.
3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.
5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.
6. Explain how image formation models help in improving image quality in applications such as autonomous driving.
7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.
8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.
9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.
10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.
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Computer Vision

CO₁ - AT₁ - Short answer

1. Sampling is the process of converting a continuous image into a digital image by dividing it into pixels. If the sampling rate is too low, important image details are lost, making the image appear blurred or distorted. This also causes aliasing, where edges become jagged. Increasing the sampling resolution captures more detail and produces a clearer image.

2. Image sensing and acquisition are the first steps in computer vision. In low-light conditions, the camera sensor receives less light, resulting in dark images with more noise and poor visibility. The quality of the sensor, lens, exposure time and lighting condition greatly influence the final digital image.

3. Aliasing occurs when an image is sampled below the required sampling frequency. As a result, straight edges appear jagged and unwanted patterns or distortions become visible.

Quantization with fewer intensity levels may also reduce image quality.

4. Pixel resolution refers to the total number of pixels in an image, while intensity resolution refers to the number of brightness or gray levels that each pixel can represent. Higher pixel resolution improves image sharpness and allows fine details to be captured.

5. A Real-time Surveillance System mainly uses high-level computer vision because it performs intelligent tasks such as detecting, recognizing and tracking people or vehicles. It continuously analyses video frames and makes decisions automatically without human intervention.

6. Image formation models explain models how light from an object passes through the camera lens and forms an image on the sensor. These models help correct lens distortion improve brightness, adjust perspective and estimate depth information.

7. Quantization converts continuous image intensity value into discrete digital values that can be stored and processed by a computer. A higher number of quantization level preserve more image details and introduces smoother brightness changes.

8. Noise is introduced during image acquisition because of poor lighting, electronic interference, sensor limitations or high camera sensitivity. It reduces image clarity and makes feature extraction more difficult.

9. Facial recognition involves all three levels of computer vision. low-level vision captures the image and removes noise, mid-level vision extracts important facial features such as the eyes, nose and mouth. High-level vision compares these features with stored database image and identifies the person.

10. when an image is captured with low resolution, fewer pixels represent the scene, making objects appear blurred and difficult to identify. Improper image formation, poor focus or inadequate lighting further reduce visibility. High-quality images are essential for obtaining accurate and reliable analysis results.